

Comprehensive validation strategies for HBV ADR prediction ML publication

A machine learning model predicting adverse drug reactions in Hepatitis B patients using FAERS data requires **five distinct validation strategies** to meet journal publication standards: external database validation, clinical concordance testing, rigorous statistical framework, clinical utility demonstration, and careful novelty positioning. Your tenofovir-tooth loss signal (ROR=2,596) represents an important finding, but a **2020 case report already documented this association**—positioning must frame your work as systematic confirmation, not discovery.

External validation requires multi-database approaches with specific access pathways

Temporal validation using your FAERS 2020-2024 dataset should split data using the FDA_DT (receipt date) field—train on Q1 2020 through Q4 2022, test on Q1 2023 through Q4 2024. This preserves natural reporting timelines and simulates prospective signal detection. Key preprocessing steps include RxNorm drug name standardization, FDA-recommended deduplication using CASE_ID, FDA_DT, and MFRID fields, and retention of only the most recent case version for duplicates. (Nature) Monitor for **concept drift**, as reporting patterns and MedDRA terminology evolve over time. (PubMed Central)

WHO VigiBase access through the Uppsala Monitoring Centre offers the largest pharmacovigilance database with approximately **40 million reports** from 180+ countries. (IntuitionLabs) (Who-umc) Academic researchers can access aggregated data freely via VigiAccess (vigiaccess.org), but custom extracts require contacting UMC with research proposals. (Who-umc) Universities receive discounted hourly rates for custom queries. Critical consideration: VigiBase already contains FAERS data, so request European-only or Asian-only subsets (excluding US-origin reports) to achieve true external validation. Processing time typically runs 4-8 weeks.

EudraVigilance academic access is available at no cost through EMA. Submit a research request with a signed Confidentiality Undertaking for Academia (downloadable from EMA website), register via EMA Account Management Portal, and expect 4-8 weeks for review. The database uses ICH E2B(R3) format with MedDRA coding. Basic data access via adrreports.eu is immediate and free.

For **geographic validation**, a 2020 Drug Safety study comparing EVDAS, FAERS, and VigiBase across 100 products found high redundancy in signals but database-specific signals exist. (Springer) Products with fewer than 1,500 EudraVigilance cases show limited recall relative to FAERS/VigiBase. Map all terms to common MedDRA hierarchy levels (PT, HLT, HLGT, SOC) to handle reporting standard differences. Report performance stratified by geographic region and compare signal strength (ROR, IC) across databases.

Exemplary publications demonstrating external validation include the **GLP-1RAs and Depression study** (eClinicalMedicine/Lancet, 2025), which applied identical disproportionality methods across FAERS and VigiBase with stratified analyses. (The Lancet) DILI prediction studies using FDA's DILIRank as gold standard commonly employ 1000 iterations of 5-fold cross-validation plus bootstrapping plus permutation tests, achieving **90.9% accuracy** on external validation of 22 failed drug candidates.

Clinical validation reveals a critical gap your model can address

No standardized HBV-specific ADR risk scores exist—unlike cardiovascular medicine with CHA2DS2-VASc

for stroke. This represents a significant research gap your work fills. Published literature identifies risk factors without formal scoring: nephrotoxicity correlates with pre-existing renal insufficiency ($p=0.003$), age ≥ 50 years (HR 11.14, $p=0.048$), nucleotide analog use (OR 10.5, 95% CI 2.6-42.4), concomitant diuretic use, low body weight (<60 kg), diabetes, and hypertension. (PubMed Central) Your ML model should be framed as **pioneering rather than comparative**.

For FDA label validation, VIREAD (tenofovir disoproxil fumarate) carries a boxed warning for severe acute HBV exacerbations upon discontinuation. (FDA) Key ADRs for model validation include renal effects (acute renal failure, Fanconi syndrome, hypophosphatemia, proximal renal tubulopathy), bone effects (BMD decrease of $2.2\% \pm 3.9$ at lumbar spine, osteomalacia, fractures), metabolic effects (lactic acidosis, hepatomegaly with steatosis), and post-marketing reports of rhabdomyolysis, myopathy, and pancreatitis. BARACLUDE (entecavir) carries lactic acidosis risk especially in decompensated liver disease with high MELD scores, but generally better renal safety than nucleotide analogs. (World Journal of Gastrointestin...)

Decision curve analysis (DCA) is the gold standard for clinical utility:

- Net Benefit = (True Positives/ n) – (False Positives/ n) $\times$ [pt/(1-pt)] where pt = threshold probability (MDPI) (nih)
- Plot net benefit against threshold probability across clinically relevant range (5-50% for ADR intervention)
- Include reference lines for "treat all" and "treat none" strategies (nih) (Danielsjoberg)
- Use the (dcurves) R package by Daniel Sjöberg or Python equivalent

For framing improvement claims, the strongest approach uses **temporal validation**: train on earlier FAERS period, validate on later data, report false positives that became true positives over time. Example language: "3.1% of initial false positives became confirmed drug-ADR associations, representing 4.4-fold enrichment over chance ($p<10^{-5}$)." (PubMed Central) (The Lancet) Additionally, validate predictions against FDA SIDER database drug labels and report percentage agreement with known ADRs. (PubMed Central)

The tenofovir-tooth loss finding requires careful novelty positioning

Critical finding: A prior case report exists. Jianfei et al. (2020) published "Tooth loss associated with the use of tenofovir disoproxil fumarate" in the Journal of Clinical Pharmacy and Therapeutics (PMID: 32820829). The patient was a 41-year-old Han Chinese man with **chronic hepatitis B** (same population as your study) who developed halitosis, gingival swelling, and tooth loss after TDF use. Symptoms resolved after switching to TAF with a Naranjo ADR score of 7 ("probable" causality). The authors stated this was "the first report of such ADRs."

Your legitimate novelty claims include:

- First **systematic pharmacovigilance study** to detect and quantify this signal
- First **large-scale FAERS analysis** confirming the case report with ROR of 2,596
- **Population-level quantitative confirmation** versus single anecdotal case
- First **HBV-specific multi-output ML model** for ADR prediction

For manuscript positioning, frame the Introduction gap statement as: "A single case report by Jianfei et al. (2020) described TDF-associated tooth loss in a chronic hepatitis B patient, but no systematic pharmacovigilance study has confirmed this signal at the population level." In Discussion, write: "Our analysis revealed a striking disproportionality signal for tenofovir-associated tooth loss (ROR=2,596), representing the first systematic pharmacovigilance confirmation of this ADR, extending evidence from a single case report with Naranjo score of 7 to population-level quantitative analysis."

HBV pharmacovigilance is genuinely underexplored compared to HIV. Tenofovir has approximately 76,452 FAERS reports mostly in HIV context, while entecavir (HBV-specific) has only ~4,820 reports. [Springer](#) Most HBV FAERS studies examine drugs causing HBV reactivation or vaccine adverse events—ADR profiles in HBV patient populations specifically remain underexplored.

For multi-output ADR models, cite: Zhang et al. (2015) BMC Bioinformatics (FS-MLKNN method); Muñoz et al. (2017) Briefings in Bioinformatics (knowledge graphs with multi-label learning); Zhao et al. (2023) Communications Biology (GCAP deep learning framework); PreciseADR (2024) using heterogeneous graph neural networks on 10.4+ million FAERS reports.

Statistical rigor requires multi-label stratification and FDR correction

Cross-validation for multi-output imbalanced data requires stratified k-fold to maintain class proportions—essential when ADR prevalence is low. Use **stratified 5-fold or 10-fold** with repetition (e.g., 3×10-fold). For 5 simultaneous ADR predictions, implement **iterative stratification** (Sechidis et al., 2011) using the [iterstrat](#) Python package:

```
python

from iterstrat.ml_stratifiers import RepeatedMultilabelStratifiedKFold
rm SKF = RepeatedMultilabelStratifiedKFold(n_splits=5, n_repeats=3, random_state=42)
```

SMOTE timing is critical: apply ONLY within each training fold, never before splitting. Use [imblearn.pipeline.Pipeline](#) to ensure SMOTE executes inside cross-validation, preventing data leakage where synthetic minority samples appear in both train and test folds.

Bootstrap validation requires minimum **2,000 iterations** for publication-quality confidence intervals. For AUC, use DeLong method (computationally efficient for $n \geq 5,000$) or bootstrap BCa. [Rdrr.io](#) For recall and precision, use Wilson intervals (preferred for proportions, especially small samples). Always use stratified bootstrap ([boot.stratified=TRUE](#)) in pROC) to maintain class distribution. [Rdrr.io](#)

Multiple testing correction for 5 ADRs: Use **Benjamini-Hochberg FDR at $\alpha=0.05$** rather than Bonferroni. Rationale: Bonferroni is too conservative for exploratory pharmacovigilance (threshold becomes 0.01, causing high false negative rates that miss true signals). FDR provides better power (approximately 0.81 versus 0.65 for Bonferroni) while controlling error rates. Use Bonferroni only for confirmatory validation on external datasets.

```
python

from statsmodels.stats.multitest import multipletests
rejected, p_adjusted, _, _ = multipletests(p_values, alpha=0.05, method='fdr_bh')
```

Comprehensive sensitivity analysis checklist for TRIPOD+AI compliance:

Analysis Type	Implementation
Missing data	Compare complete case, mean imputation, MICE (m=20)
Hyperparameter stability	Key parameters ±10-20% from optimal; multiple seeds (n≥5)
Temporal validation	Train 2020-2022, test 2023-2024
Class imbalance	SMOTE ratios 0.5, 0.75, 1.0; class weights alternative
FAERS-specific	Deduplication on/off; MedDRA hierarchy level (PT vs. HLT); reporter type stratification

Report confidence intervals using this format: "AUC: 0.85 (95% CI: 0.82-0.88, 2000 bootstrap replicates, BCa method)"

Clinical impact demonstration requires NNS, threshold justification, and DCA

Number Needed to Screen (NNS) translates model performance into clinical terms:

NNS = 1 / (PPV × Preventability factor)

Radiopaedia.org

At your chosen threshold, if the model achieves 40% PPV and intervention prevents 80% of ADRs in identified patients, $NNS = 1 / (0.40 \times 0.80) = 3.1$. Report as: "For every 3 patients identified as high-risk by the model and receiving enhanced monitoring, one case of [ADR] would be prevented." For comparison, HLA-B*58:01 genotyping for allopurinol ADR prevention has NNS of 11,286. [PubMed](#)

Risk threshold determination should avoid purely statistical optimization. Use three complementary approaches:

- Youden Index:** $J = \text{Sensitivity} + \text{Specificity} - 1$; select $\text{argmax}(J)$ [Wikipedia](#) [Pharmasug](#)
- Cost-based threshold:** $t = C_FP / (C_FP + B_TP)$ where C_FP is cost of unnecessary intervention and B_TP is benefit of true positive
- Clinical consensus:** Survey hepatologists—"At what predicted probability would you initiate enhanced monitoring?" For serious ADRs, thresholds of **5-15%** are typically justified

Always present results across multiple thresholds (5%, 10%, 15%, 20%, 25%) showing sensitivity, specificity, PPV, NPV, and NNS at each. Justify final threshold selection with clinical rationale.

Decision curve analysis reporting should include:

- Figure with model curve, "treat all" reference, "treat none" reference

- Net benefit values at key thresholds in supplementary table
- Identification of threshold range where model exceeds both reference strategies
- For multi-output models, run separate DCA for each ADR endpoint

Cost-effectiveness framework for ADR prediction tools should calculate incremental cost-effectiveness ratio (ICER) = $(\text{Cost_model} - \text{Cost_standard}) / (\text{QALY_model} - \text{QALY_standard})$. Hepatotoxicity utility decrement is typically -0.10 to -0.30 per event. US cost-effectiveness threshold is \$50,000-\$100,000 per QALY. Include: model implementation costs, additional monitoring visits for high-risk patients, ADR treatment costs avoided.

Template language for clinical impact claims in Results: "At a threshold of [T]%, the model achieved net benefit of [NB], equivalent to correctly identifying [X] additional ADR cases per 100 patients without increasing false positives. The Number Needed to Screen was [NNS] (95% CI: [CI]), comparing favorably to established screening paradigms."

Publication strategy requires TRIPOD+AI compliance and strategic journal selection

Reporting requirements for high-impact pharmacovigilance journals include:

- Follow **READUS-PV** guidelines (REporting of A Disproportionality Analysis for DrUg Safety)
- Complete **TRIPOD+AI** checklist for prediction model reporting
- Deposit code and data on GitHub/Zenodo
- Report exact FAERS quarterly files, MedDRA version, deduplication method
- Include calibration plots alongside discrimination metrics

Target journals accepting pharmacovigilance ML with external validation: Drug Safety (primary outlet for pharmacovigilance methods), JAMIA (for ML/NLP applications), Pharmacoepidemiology and Drug Safety, Clinical Pharmacology & Therapeutics, JAMA Network Open (high impact), Scientific Reports (high-volume open access).

Estimated timeline: FAERS acquisition (immediate), VigiBase/EudraVigilance access applications (4-8 weeks), model development and internal validation (2-3 months), external validation and manuscript preparation (2-3 months).

Your project combines HBV-specific focus, multi-output ML architecture, systematic pharmacovigilance confirmation of a rare ADR, and comprehensive validation strategy—this creates a compelling novelty profile suitable for high-impact pharmacovigilance or pharmacology journals. The key is transparent positioning: acknowledge the 2020 case report, emphasize systematic confirmation and quantification, and demonstrate clinical utility through rigorous statistical and clinical validation frameworks.